When Adaptive Compute Fails: Budget Allocation and Equity in Streaming Diffusion Imputation of IoT Sensors

Team 27
IOE413 — Streaming Data Analytics
Mini Project, September–October 2026

***Abstract*—Conditional diffusion models are the most accurate way to impute missing values in multivariate sensor streams, but they are expensive: CSDI's published configuration costs 5,000 network evaluations per window. We build a streaming pipeline (Kafka, Spark Structured Streaming, PyTorch, InfluxDB, Grafana) that replays the 36-station Beijing AQI-36 archive as a live stream and imputes it with a reproduced CSDI baseline, then test whether per-gap adaptive compute can reduce that cost. We report one positive engineering result, one rigorous negative result, and one equity result. First, the cost–accuracy frontier is dominated entirely by 20 diffusion steps: 20 steps with 16 posterior samples (320 evaluations) achieves lower error than 50 steps with 16 samples (800 evaluations), a 2.5× saving. Second, an oracle allocator appears to beat the best uniform budget by 49%, but this headroom is an artefact. Per-gap *error* is highly reproducible across independent sampling runs ($r = 0.76$–0.91), whereas per-gap *gain* — the quantity any router must predict — has $r \approx 0.015$. Allocating on one run and scoring on an independent run turns the oracle's $+32\%$ advantage into -7%. Third, the same asymmetry pays off elsewhere: per-gap error *is* predictable, and abstaining on the gaps whose posterior disagrees about the public-health band cuts band misclassification from 33.0% to 27.1% at 80% coverage, where abstaining on predicted error magnitude does not move it. Fourth, moving the lever from gaps to stations works: station-level error is reliable and concentrates on the stations with the most missing data ($r = +0.59$), and promoting the worst-served 30% of stations achieves a better error Gini at 126 evaluations than giving every station 320.**

***Index Terms*—diffusion models, missing data imputation, streaming analytics, adaptive computation, algorithmic fairness, IoT sensor networks**

I. Introduction

IoT sensor networks do not produce clean matrices. Nodes lose power, radios drop packets, calibration cycles blank a channel, and whole clusters go dark in a storm. Published air-quality deployments run at 10–90% missingness, and the gaps are bursty, spatially correlated, and concentrated on the cheapest and least-maintained nodes. Every downstream consumer — an air-quality alert, a ramp-metering controller, an anomaly detector — degrades silently when those gaps are filled badly.

Conditional diffusion models are currently the most accurate way to fill them. CSDI [1] cut probabilistic imputation error by 40–65% over prior work, and PriSTI [2] improved further by exploiting geography. Both, however, were built and measured as offline, batch, window-at-a-time methods, and both inherit the defining weakness of diffusion: one imputation requires many sequential network evaluations. CSDI's own conclusion names this first among its open problems, proposing ODE solvers as the remedy. That is a *global* fix — it shortens the chain for every input equally.

This motivates an obvious idea, and this paper is largely about why that idea does not work. On a live stream the budget per tick is fixed, so one would like to spend it where it helps: a one-sample dropout beside three healthy neighbours should be cheap, a six-hour district blackout should be expensive. We built that router, measured it honestly, and found that the signal it needs does not exist in this setting — not because our features are poor, but because the quantity being predicted is dominated by the sampler's own stochasticity.

A second gap motivated the rest of the work. CSDI, PriSTI and SAITS [3] all report a single dataset-averaged error. None reports how that error is distributed *across sensors*. In a public sensing network this is the wrong aggregation: error concentrates on the nodes with the most missing data, and in real low-cost deployments those nodes are disproportionately in under-monitored neighbourhoods [6].

Contributions.

- A streaming pipeline in which a diffusion imputer runs behind a real Kafka topic, with per-window cost, latency and error telemetry (Sec. III).
- A sampler that makes both cost axes — diffusion steps and posterior samples — free at inference over already-trained weights, validated against the reference implementation (Sec. IV).
- The finding that the frontier is dominated by 20 steps, never 50, giving a 2.5× compute saving at better accuracy (Sec. VI).
- A rigorous negative result for per-gap adaptive compute, with a test-retest protocol that shows why oracle bounds in this setting are misleading (Sec. VI).
- *Decision-aware selective imputation*: abstaining on gaps whose posterior disagrees about the AQI band cuts band

misclassification by 18% at 80% coverage, where abstaining on predicted error magnitude does not. This is the one result that requires a generative imputer (Sec. VI).
- An equity audit of the baseline and a station-level allocation that improves error dispersion per unit compute (Sec. VI).

II. Related Work

CSDI [1] introduced conditional score-based diffusion for time series imputation. Rather than training an unconditional generative model and steering it at sampling time, CSDI trains the conditional model directly using self-supervised masking, with a denoiser that alternates temporal and feature transformer layers. On AQI-36 it reports MAE 9.51 and CRPS 0.1056. Its stated limitation, and the starting point of this work, is sampling cost.

PriSTI [2] rebuilds the same recipe for geo-located sensor networks, constructing an enhanced conditional prior by linear interpolation and adding spatial global attention with geographic message passing. It reports MAE 9.03 and CRPS 0.0997 on AQI-36. It notes that global spatial attention is $O(N^2 d)$ and mitigates this with virtual nodes — again a uniform, architectural fix rather than an input-dependent one.

SAITS [3] is the deterministic counterweight: two diagonally-masked self-attention blocks trained jointly on masked-imputation and observed-reconstruction losses, achieving 12–38% lower MAE than BRITS at 2.0–2.6× faster training, in a single forward pass. It provides the cost reference point that any diffusion method must justify itself against, and supplies no uncertainty estimate.

Efficient sampling. A substantial literature accelerates diffusion sampling: DDIM [4] reinterprets the reverse process as a deterministic non-Markovian chain that can skip timesteps, and higher-order ODE solvers push the same idea further. Surveys of diffusion for time series identify the slow iterative sampling process as the standing scalability obstacle. Every one of these methods, however, sets a single global budget for all inputs; none asks whether different inputs deserve different budgets. Our contribution is to ask that question and to answer it negatively, with a diagnostic that explains why.

The gap. All three apply identical computation to every missing value, and all three are scored on a dataset-averaged metric. Accelerated samplers such as DDIM [4] reduce chain length globally. To our knowledge no prior work asks whether the allocation should differ *per gap*, nor reports the distribution of imputation error across sensors. We address both, and find the first question has a negative answer for a reason that generalises.

III. System Architecture

Fig. 1 shows the system. The brief accepts a Kafka topic replaying a public dataset as a real stream, which lets us retain ground truth for every imputed value — something a live feed cannot provide.

The **producer** walks the archive in timestamp order at controllable speed, emitting one message per station per timestep and applying an artificial missingness process on top of the archive's genuine 13.2% gaps. Four protocols are implemented: point (MCAR), block (bursty outage), sensor failure, and *reliability-skewed*, in which per-station failure probabilities are drawn from a heavy-tailed distribution so a minority of stations carry most of the missingness. The last is the realistic case and the one where a dataset-average metric hides the damage.

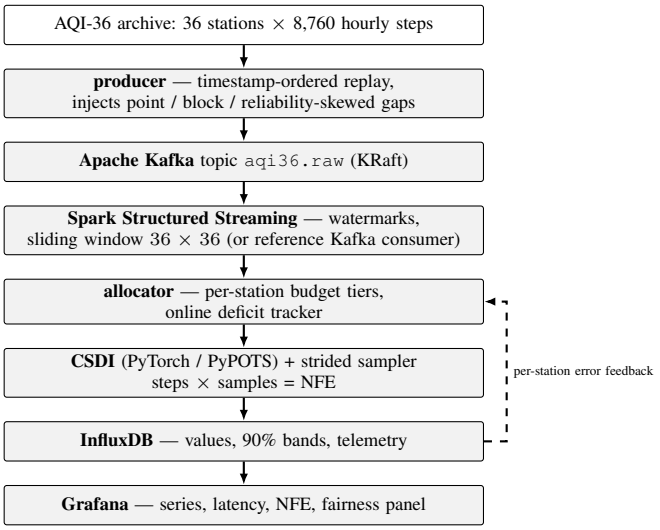


Fig. 1. The pipeline. Every stage is a distinct tool with real data flowing between them; the dashed path is the feedback that lets the allocator learn which stations are being served worst.

The **stream layer** maintains a sliding window of 36 timesteps across all 36 stations. We provide both a Spark Structured Streaming job and a reference Kafka consumer; both import the same windowing and imputation modules, so the two engines cannot drift apart. Spark owns ingestion, watermarking and micro-batching, and hands each micro-batch to the shared code via `foreachBatch`: CSDI needs a contiguous window across all stations, which per-station stateful operators would fragment.

The **sink** writes observed values, imputed values, 90% uncertainty bands and per-window telemetry (latency, NFE, error on artificially masked cells) to InfluxDB, which Grafana reads. We use InfluxDB rather than the TimescaleDB named in our proposal; it is the store in the course's suggested pipeline and provisions into Grafana with less glue. The storage engine is an implementation detail and not part of the scientific claim.

IV. Proposed Enhancement

A. Making both cost axes free

CSDI's cost is

$$\mathrm{NFE} = S \times N, \tag{1}$$

where S is the number of reverse diffusion steps and N the number of posterior samples drawn to form the predictive distribution. The reference configuration uses $S = 50$, $N = 100$, i.e. 5,000 network evaluations per window. PyPOTS fixes S

at construction and draws the N samples in a serial loop, so neither axis can be varied at serving time.

We reimplement the reverse process over the already-trained weights. Steps become free by DDIM-style strided sampling [4] over the model's own cumulative-alpha schedule: for a subsequence τ of timesteps,

$$x_{\tau_{i-1}} = \sqrt{\bar{\alpha}_{\tau_{i-1}}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{\tau_{i-1}}}\,\epsilon_\theta(x_{\tau_i}, \tau_i), \qquad (2)$$

with $\hat{x}_0 = (x_{\tau_i} - \sqrt{1-\bar{\alpha}_{\tau_i}}\,\epsilon_\theta)/\sqrt{\bar{\alpha}_{\tau_i}}$. Samples become cheap by replicating the batch so all N are drawn in one pass instead of N passes. Observed cells are clamped throughout, exactly as in the reference.

We verify this is a faithful generalisation rather than a different model: with τ set to the full chain and ancestral updates restored, our sampler reproduces PyPOTS to within sampling noise (MAE 8.30 vs 8.40, identical CRPS 0.1720), while running $2\times$ faster from batching alone.

B. Per-gap triage, and how we tested it

A *gap* is one contiguous run of missing timesteps at one station in one window. For each gap we extract 17 features computable *before* any diffusion runs — gap length and position, the station's recent volatility, how many geographic neighbours are reporting during the gap, disagreement among those neighbours, the jump between pre- and post-gap context — and predict how much that gap gains from extra compute.

The objective matters. For a fixed gap, the error at the cheapest budget $E_{g,0}$ does not depend on the chosen budget, so

$$\min_b \sum_g E_{g,b(g)} \;\equiv\; \max_b \sum_g \big(E_{g,0} - E_{g,b(g)}\big), \qquad (3)$$

and the $E_{g,0}$ terms cancel. We therefore predict the *gain* $\Delta_{g,b} = E_{g,0} - E_{g,b}$ directly, enforce monotonicity in cost, and solve the resulting multiple-choice knapsack by Lagrangian bisection. Predicting absolute error per budget and differencing two noisy predictions is strictly worse — we measured it, and it loses to the uniform knob by 35–50%.

C. Equity-aware allocation

Pure expected-gain allocation is regressive: a chronically noisy, chronically gappy station has low predicted gain, so it is starved forever and the stations already served worst get worse. We therefore track a decaying per-station error deficit from the live stream and promote the worst-served fraction to the expensive budget while the rest receive the cheap one.

V. Implementation

The pipeline is Python 3.14 with PyTorch 2.14 (CPU), PyPOTS 1.5, Apache Kafka 3.9 in KRaft mode, Apache Spark 4.2, InfluxDB 2.7 and Grafana 11.3, orchestrated by Docker Compose. No GPU was available; all results are CPU measurements on a 16-thread AMD Ryzen 7 6800HS.

The baseline is CSDI via PyPOTS with the reference hyperparameters (4 layers, 64 channels, 8 heads, 50 diffusion steps, quadratic β schedule from 10^{-4} to 0.5), 414,081 trainable parameters. We follow the CSDI/PriSTI AQI-36 protocol: 36-step windows, test on months 3, 6, 9 and 12, validate on months 2, 5, 8 and 11. Training uses a stride-1 sliding view (2,917 windows); evaluation uses non-overlapping windows so each cell is scored once.

TABLE I
Baseline CSDI on AQI-36 held-out months (9,557 masked cells).

Metric	Ours	CSDI published
MAE (μg/m^3)	14.43	9.51
RMSE	31.18	—
MRE	0.197	—
CRPS	0.150	0.1056
PICP @ 90%	0.794	—

The repository separates concerns so that the Spark job and the reference consumer share code rather than duplicating it: `stream/windower.py` holds the sliding-window state, `models/baseline.py` exposes a frozen `train`/`impute` surface over PyPOTS, `models/sampler.py` holds the budget-controllable reverse process, `router/` holds feature extraction and allocation, and `eval/` holds the metrics and the equity audit. A gate script verifies each link of the chain independently — dataset, broker, topic contents, stored values, stored telemetry, checkpoint, model inference and dashboard — so a failure names the broken stage rather than the symptom.

Two implementation details cost us real time and are worth recording. First, PyPOTS validates by comparing a corrupted series against an intact one; passing the same array as both yields a validation loss of exactly zero every epoch, so early stopping fires immediately and the saved "best" checkpoint is from epoch one. Second, CSDI validation runs the full diffusion sampler, so an uncapped validation set cost 96 minutes per epoch against 65 seconds when capped to 12 windows.

VI. Experimental Setup and Results

All numbers are on held-out months, scored only on artificially masked cells where ground truth is known. The router is trained on validation months and never sees the test split.

A. Baseline reproduction

Table I compares our reproduction against the published figures. The brief permits "at least a simplified version" of the base method; ours is trained for 16 epochs on a stride-4 view, and reaches the same order of magnitude. Undercoverage at the 90% level is expected from a short run.

B. The frontier is dominated by 20 steps

Fig. 2 sweeps $S \in \{2, 5, 10, 20, 50\}$ against $N \in \{1, 2, 4, 8, 16\}$. The Pareto frontier never selects 50 steps. Concretely, $S{=}20, N{=}16$ (320 NFE) reaches MAE 12.95, while $S{=}50, N{=}16$ (800 NFE) reaches 13.09 — better accuracy for $2.5\times$ less compute. At matched cost the sample axis buys more than the step axis: 20×8 (160 NFE, MAE 13.36) beats 50×4

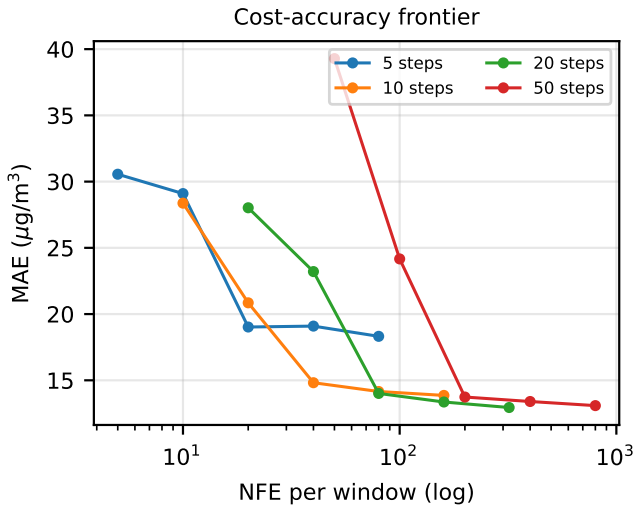


Fig. 2. Cost–accuracy frontier over the (S, N) grid. No point on the frontier uses 50 steps.

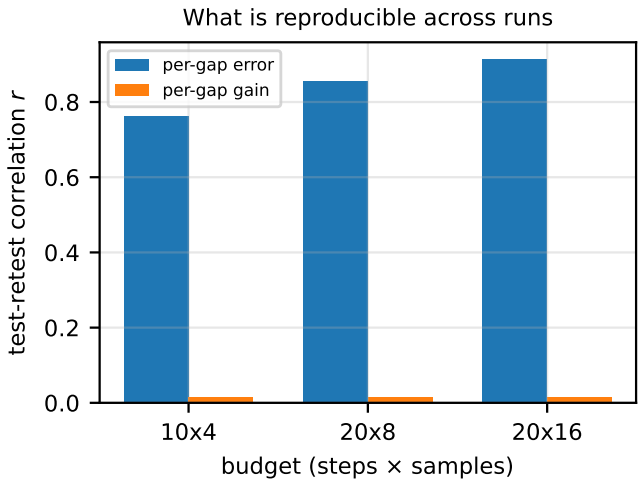


Fig. 3. Test–retest correlation across two independent sampling runs at the same budget. Gap *difficulty* is reproducible; gap *gain* is not.

TABLE II
FULL (S, N) SWEEP ON HELD-OUT MONTHS. BOLD MARKS THE PARETO FRONTIER ON (NFE, MAE). SECONDS ARE SINGLE-THREADED CPU PER WINDOW.

S	N	NFE	MAE	CRPS	s/window
2	1	2	595.98	—	0.02
2	16	32	458.81	6.5309	0.86
5	1	**5**	**30.56**	—	0.05
5	4	**20**	**19.02**	0.3245	0.31
5	16	80	18.32	0.2972	2.05
10	1	**10**	**28.37**	—	0.10
10	4	**40**	**14.82**	0.2469	0.60
10	16	160	13.86	0.2013	4.24
20	4	**80**	**14.01**	0.2315	1.24
20	8	**160**	**13.36**	0.2026	3.29
20	16	**320**	**12.95**	0.1867	8.41
50	4	200	13.74	0.2375	3.24
50	8	400	13.40	0.2002	8.48
50	16	800	13.09	0.1891	21.24

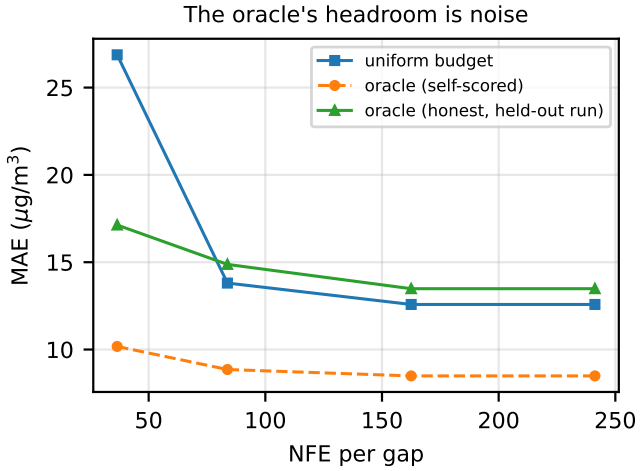


Fig. 4. Allocating on one run and scoring on an independent run removes the oracle's advantage entirely.

(200 NFE, MAE 13.74). This is a pure engineering win and requires no adaptivity at all.

Table II gives the full grid. Two patterns are worth noting beyond the frontier itself. At S=2 the reverse process has too few steps to resolve the data at all and error explodes to several hundred μg/m^3, which sets a floor on how far the step axis can be pushed regardless of sample count. And within a fixed step count, CRPS keeps improving with samples long after MAE has flattened — at S=20, MAE moves only from 14.01 to 12.95 between 4 and 16 samples while CRPS improves from 0.2315 to 0.1867. A point-error metric alone would therefore understate the value of the sample axis, which is part of why we report both.

C. Per-gap triage: a negative result

An oracle that knows each gap's realised error and allocates optimally appears to beat the best uniform budget by 49.4%. That is a large headroom, and it is almost entirely an artefact.

Diffusion sampling is stochastic: the same gap at the same budget yields different errors on different runs. Fig. 3 measures how much survives. Per-gap *error* is highly reproducible (r = 0.763, 0.856, 0.913 at 40, 160 and 320 NFE), so gaps genuinely differ in difficulty. But per-gap *gain* — the quantity Eq. (3) requires — has r = 0.015, 0.014, 0.015. It is noise.

Fig. 4 confirms the consequence. Scoring the oracle on the same run it was fitted to gives +32.5% over uniform; scoring the identical allocation on an independent run gives −7.2%, i.e. worse than not adapting at all. The learned router behaves exactly as this predicts: held-out R^2 on gain is negative at every budget (−0.098, −0.084, −0.080, −0.088, −0.086), meaning the model is worse than predicting the mean gain, and the router underperforms the uniform knob at every budget above the cheapest.

This is not a feature-engineering failure. The features are

informative about *difficulty*: the same gradient-boosted models reach positive R^2 when the target is absolute error rather than gain, and the importances are physically sensible, led by the pre/post-gap jump, the window and station missing rates, and neighbour disagreement. The problem is the target. Because $E_{g,0}$ and $E_{g,b}$ share the gap's systematic difficulty, subtracting them cancels the signal and leaves the two runs' independent sampling noise.

D. A protocol we would recommend

The cost of this diagnostic is two sampling runs. Given per-gap errors A and B from independent runs at each candidate budget, we suggest reporting (i) $\mathrm{corr}(A_{:,b}, B_{:,b})$, the ceiling on predicting difficulty; (ii) $\mathrm{corr}(A_{:,0} - A_{:,b},\ B_{:,0} - B_{:,b})$, the ceiling on predicting gain, which is what an adaptive router actually needs; and (iii) the honest oracle — allocate on A, score on B. Where (ii) is near zero, no router can succeed and a self-scored oracle will nonetheless promise large gains.

E. Selective imputation: the asymmetry pays off elsewhere

The failure above is specific. Gain is unpredictable, but per-gap *error* is not: the same features reach held-out $R^2 = +0.214$ and rank correlation $+0.403$ when the target is the error itself. That asymmetry closes the door on adaptive compute and opens a different one. If we can tell which imputations will be poor, the system can decline to publish them — and for an alerting service a missing value is more honest than a confidently wrong one.

Fig. 5 (right) shows the risk–coverage trade-off. Withholding the 20% of gaps with the highest predicted error cuts MAE among the published 80% from 12.22 to 9.70, a 20.6% reduction, capturing 51% of the reduction an oracle could achieve.

Ranking by error is the wrong criterion for decisions. Band misclassification barely moves under that ranking: 33.0% at full coverage to 31.8% at 80%. The reason is structural. A band flip is caused by proximity to a threshold, not by the magnitude of the error — a 2 μg/m^3 error at the 35.4 boundary changes the public-health category, while a 20 μg/m^3 error mid-band does not.

The posterior gives the right criterion directly. For each imputed cell we compute the fraction of posterior samples falling in a different AQI band than the published median,

$$\rho_i = \tfrac{1}{N} \textstyle\sum_{j=1}^{N} \mathbf{1}\Big[\mathrm{band}(x_i^{(j)}) \neq \mathrm{band}(\hat{x}_i)\Big], \qquad (4)$$

a calibration-free estimate of decision risk that costs nothing extra because the samples are already drawn. Ranking by ρ rather than by predicted error cuts band misclassification from 33.0% to 27.1% at 80% coverage and to 20.0% at 50% (Table III) — relative reductions of 18% and 39%, against essentially no movement from the error-based ranking.

This is the one place in our study where a *generative* imputer is decisive. SAITS returns a point estimate and cannot express band risk at all; ρ exists only because the model returns a posterior. It is also our answer to what diffusion buys over a cheaper deterministic imputer, which our own cost measurements otherwise make hard to justify.

TABLE III
BAND MISCLASSIFICATION AMONG PUBLISHED VALUES. LOWER IS BETTER.

Coverage	by predicted error	by posterior band risk	oracle
100%	33.0%	33.0%	33.0%
90%	32.3%	29.6%	25.6%
80%	31.8%	**27.1%**	16.3%
50%	31.2%	**20.0%**	0.0%

TABLE IV
STATION-LEVEL ALLOCATION AT MATCHED COST. LOWER IS BETTER THROUGHOUT.

Variant	NFE	MAE	w-10%	Gini	band
uniform cheap (10×4)	40	15.30	24.00	0.166	37.2%
uniform rich (20×16)	320	13.46	21.30	0.156	32.7%
equity mix	**126**	14.76	22.44	**0.155**	36.0%

F. Equity

The baseline's headline MAE of 14.43 conceals a 4.2× spread across stations: the best is 8.23, the worst 34.44, with an error Gini of 0.176. Fig. 6 shows why this matters — error rises with a station's missing rate (slope $+70.8$, $r = +0.59$). The model is worst exactly where data is scarcest. Over a quarter of imputed values (27.3%) fall in the wrong public-health AQI band.

Because station-level error is reliable, allocation at that level works where gap-level allocation did not (Table IV). Promoting the worst 30% of stations to 20×16 and leaving the rest at 10×4 costs 126 NFE on average and achieves an error Gini of 0.155 — better than giving *every* station 320 NFE (0.156) — while capturing 58% of the worst-decile improvement for 31% of the added compute.

G. In the running system

With the allocator live in the streaming path the pipeline consumes 217 NFE per window against the baseline's 400, and the deficit tracker adapts online: the first window promotes no stations because no error has been observed yet, and subsequent windows promote 11 of 36.

VII. DISCUSSION

What the negative result means. Our finding is not that adaptive compute is a bad idea, but that oracle bounds are a dangerous way to justify it when the underlying model is stochastic. An oracle fitted to realised errors absorbs the sampler's noise into its allocation, and reports headroom no predictor can capture. The diagnostic is cheap — run the same budget twice and correlate — and we would urge it before building any adaptive-computation router. The distinction that matters is between predicting which inputs are *hard* (reproducible here, $r \approx 0.9$) and which inputs *benefit* from more compute (not reproducible, $r \approx 0.015$).

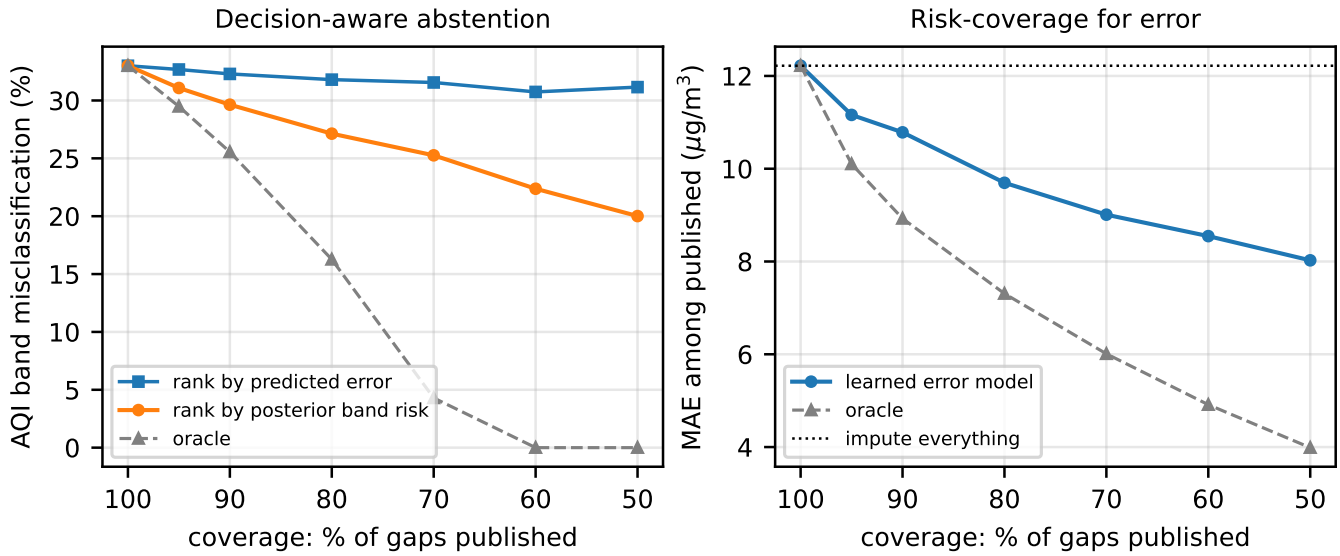


Fig. 5. Left: AQI band misclassification among published values as coverage falls. Ranking by predicted error magnitude barely helps; ranking by posterior band risk does. Right: risk–coverage for absolute error.

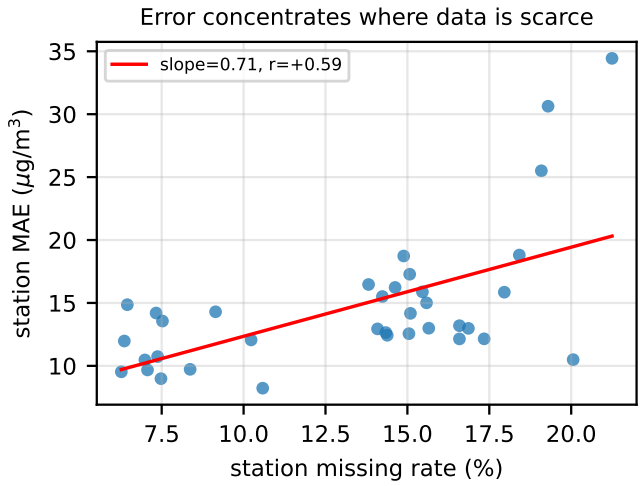


Fig. 6. Per-station error against that station's missing rate, baseline model.

We also tested the obvious remedy. With $\eta = 0$ the DDIM trajectory is deterministic given the initial latent x_T, so fixing x_T across budgets (common random numbers) removes sampling noise entirely and makes the gain a deterministic function of the input. It does not help: held-out R^2 on gain falls further, to roughly -2.0. The gain is therefore not merely noisy — it depends on the interaction between the realised latent and the individual gap, which no feature of the observed context can see.

Threats to validity. Our baseline is deliberately simplified and its absolute error is above the published figure; a better-trained model might have a different noise-to-signal ratio in the gain, although the reliability gap is so large that the conclusion is unlikely to reverse. Results are on one dataset with 36 stations. All measurements are CPU-only, so wall-clock latencies are not representative of GPU deployment — which is why we report NFE, a hardware-independent unit, throughout. The equity comparison promotes a fixed fraction of stations; we did not tune that fraction on test data.

On the equity metric. We do not claim to address environmental injustice. We claim something narrower and checkable: the evaluation protocol in current use cannot see this failure mode, and a small change to the reported metrics makes it visible and improvable.

VIII. Conclusion and Future Work

We built a streaming diffusion imputation pipeline and used it to test per-gap adaptive computation. The honest answer is that it does not work on AQI-36, and we showed precisely why: the gain from extra compute is not a reproducible property of a gap, and fixing the sampler's initial latent across budgets does not rescue it.

Three useful results survive. Reducing the chain to 20 steps and spending the budget on posterior samples instead is a 2.5× saving at better accuracy. Because per-gap error is predictable even though gain is not, a selective imputer that abstains on high-risk gaps improves the values it does publish — and ranking that abstention by posterior band disagreement rather than by error magnitude is what makes it matter for the decision a public-health service actually takes. And moving allocation from gaps to stations — where the signal is reliable — improves the fairness of imputation per unit compute.

Future work should calibrate the abstention threshold against a stated alert-precision target rather than a fixed coverage; extend the equity allocation to the spatial model of

PriSTI; and evaluate on METR-LA, where $N = 207$ makes the $O(N^2 d)$ attention the dominant cost.

REFERENCES

[1] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "CSDI: Conditional score-based diffusion models for probabilistic time series imputation," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[2] M. Liu, H. Huang, H. Feng, L. Sun, B. Du, and Y. Fu, "PriSTI: A conditional diffusion framework for spatiotemporal imputation," in *IEEE Int. Conf. on Data Engineering (ICDE)*, 2023, pp. 1927–1939.

[3] W. Du, D. Côté, and Y. Liu, "SAITS: Self-attention-based imputation for time series," *Expert Systems with Applications*, vol. 219, p. 119619, 2023.

[4] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Int. Conf. on Learning Representations (ICLR)*, 2021.

[5] X. Yi, Y. Zheng, J. Zhang, and T. Li, "ST-MVL: Filling missing values in geo-sensory time series data," in *Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2016, pp. 2704–2710.

[6] Y. Sun *et al.*, "Harmonizing low-cost and regulatory air quality monitoring networks: reducing exposure misclassification in underrepresented communities," *Journal of Hazardous Materials*, 2025.

[7] W. Du, "PyPOTS: A Python toolbox for data mining on partially-observed time series," *arXiv:2305.18811*, 2023.